Started on	Monday, 23 November 2020, 14:10
State	Finished
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Time taken	3 hours 4 mins

Information

For the following three questions you will have to submit code parts and the eventual binary program for a simple calculator that uses the emulator UART for input and output.

Please use the latest version of the emulator, available on SunLearn - version 0.10.

The input that your program must be able to handle will be an ASCII string of the form

```
(-)<num1>[+,-,x,/]<num2>=
```

- The initial '-' character is optional, but can be present to indicate that the first number in the calculation is negative.
- <num1> and <num2> are placeholders. Both are positive integers in the range 0..9999.
- The calculation operator is either addition '+', subtraction '-', multiplication 'x' or division '/'.
- The input will end with an equals character, '='. At this point your calculator should be triggered to calculate the output and send the result back to the UART. The output should be the ASCII representation of the calculated result, in the form

```
(-)<result>\n
```

- The initial '-' sign is optionally output if the calculated result is negative.
- <result> should be the integer result of the calculation.
- The output must have a newline character (\n) at the end
- If the requested operation is division, and <num2> equals zero, the program must reply with 'divide by zero\n' instead of the above.

Your program must be able to repeatedly capture input and perform calculations. Every time a '=' character is received, it should process the preceding input and output the answer. You should also handle invalid input, i.e. input that is not in the format prescribed above. In such a case, it should output the text message "invalid input\n". (note the lower case, and newline character at the end of the message.)

Vir die volgende drie vrae moet jy dele bronkode en die vertaalde (binêere) program indien vir 'n eenvoudige sakrekenaar wat die emuleerder UART gebruik vir intree en uittree.

Gebruik asb. die mees onlangste weergawe van die emuleerder op SunLearn - weergawe 0.10.

Die intree wat jou program moet kan hanteer is 'n ASCII karakter string van die vorm

```
(-)<num1>[+,-,x,/]<num2>=
```

- *Die aanvanklike '-' karakter is opsioneel en kan voorkom indien die eerste getal in die bewerking negatief is*
- *Die <num1> en <num2> plekhouers is beide positiewe heelgetalle in die reeks 0..9999*
- *Die bewerking kan een van die volgende wees: '+' vir optel, '-' vir afrek, 'x' vir vermenigvuldiging en '-' vir deel*
- *Die intree sal eindig met 'n gelykaan karakter, '='. As hierdie karakter ontvang word moet jou program die berekening uittree op die voorafgaande uittree en die uittree uitstuur op die UART. Die uittree moet 'n ASCII voorstelling wees van die berekeningsresultaat, in die vorm:*

```
(-)<result>\n
```

- *Die aanvanklike '-' karakter moet opsioneel gestuur word indien die resultaat van die bewerking negatief is*
- *<result> moet die heelgetal resultaat van die bewerking wees*
- *Die uittree moet 'n nuwe-lyn '\n' karakter aan die einde hê.*
- *Indien die bewerking wat uitgevoer moet word 'n deel bewerking is en <num2> is nul, moet die program 'n boodskap 'divide by zero\n' boodskap terugvoer in plaas van die bg. terugkeer.*

Jou program moet herhaalde intree kan hanteer en terugvoer gee elke keer wat 'n '=' karakter voorkom. Jy moet ook ongeldige intree hanteer. Indien die intree nie voldoen aan die formaat wat hierbo beskryf word nie, moet dit terugkeer met 'n 'invalid input\n' boodskap op die UART. Let op die klein letters en die nuwe-lyn karakter aan die einde.

Examples of input/output is shown in the table: / *Voorbeelde van intree en uittree word in die tabel hieronder gegee:*

Input	-123+456=
Output	333
Input	12+34=
Output	46
Input	-1234*9876=
Output	-12186984
Input	15/16=
Output	0

Input	123/0=
Output	divide by zero
Input	aaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa=
Output	invalid input
Input	1234567x9876543=
Output	invalid input

Your program has to implement the following two functions (you will be asked to submit them independently in the next questions)

Jou program moet die volgende twee funksies implementeer (jy sal hulle apart moet indien in die opeenvolgende vrae).

1. receive_byte :

```
/**
 * @brief Add a received byte to the input buffer, and test for terminating character
 *
 * This function will add the byte that is passed in the @rxbyte parameter to the
 * input buffer. The @inputbuffer is supplied as parameter to the function, as
 * well as a pointer to the position where the next byte must be placed in @inputbuffer.
 * the write position is passed as a pointer argument, @ptr_writepos, because the
 * function has to increment the write position after adding the byte to the input buffer,
 * and return it to the calling program.
 * The function must only allow writing into the buffer if the value pointed to by
 * @ptr_writepos is less than 11, because 11 is the maximum length of valid input
 * the function returns 1 if the received byte is the terminating character, otherwise
 * it returns 0.
 *
 * @param inputbuffer The array into which the received byte must be written
 * @param ptr_writepos Pointer to the position (index) where the received byte must
 *                     be written.
 * @param rxbyte The received byte
 *
 * @return 1 if a terminating characted was received. 0 otherwise
 */
```

```
uint8_t receive_byte(uint8_t* inputbuffer, uint8_t* ptr_writepos, uint8_t rxbyte)
{
   // pseudocode of what must be implemented:
   // if writepos < 11
   //    . write rxbyte into inputbuffer at index writepos
   //    . increment writepos
   //
   // if rxbyte is a '='
   //    return 1
   // else
   //    return 0
}
```

2. calc_result

```
/**
 * @brief Calculate the result for the provided arguments and opcode
 *
 * This function will perform the requested calculation and return the result
 *
 * @param num1 First number to use in the calculation.
 * @param num2 Second number to use in the calculation. If opcode equals 4 (division)
 *             then this parameter may not be 0.
 * @param opcode 1 = addition, 2 = subtraction, 3 = multiplication, 4 = division,
 *               for all other values of opcode, return zero
 *
 * @return The calculation result
 */
```

```
int32_t calc_result(int16_t num1, int16_t num2, uint8_t opcode);
```

```

```

Example usage for the two functions. Note that the `process_input` function is explained later on. / *Voorbeeld gebruik van die bg. two funksies. Let wel dat die `process_input` funksie verder ondertoe beskryf word.*

```
volatile uint8_t char_received; // flag to indicate that a character was received from the UART
volatile uint8_t received_byte; // the byte that was received from the UART
uint8_t thebuffer[11];          // the buffer to hold the UART input
uint8_t writepos = 0;           // index where next UART received byte will be written
int16_t num1;                   // first number extracted from input buffer
int16_t num2;                   // second number extracted from input buffer
uint8_t opcode;                 // numeric code op operation to perform (1 = addition, 2 = subtraction, 3 =
multiplication, 4 = division)
int32_t result;                 // result of calculation

...
while (1)
{
   if (char_received)
   {
      if (receive_byte(thebuffer, &writepos, received_byte) == 1)
      {
          if (process_input(thebuffer, &num1, &num2, &opcode) == 1)
          {
              if ((opcode == 4) && (num2 == 0))
              {
                  // reply to UART with 'divide by zero'
              }
              else
              {
                  result = calc_result(num1, num2, opcode);
                  // output result to UART
              }
          }
          else
          {
              // reply to UART with 'invalid input'
          }
          writepos = 0; // reset write index, ready for next input
      }
      char_received = 0;
   }
}
```

The following two submissions will require you to submit the `receive_byte` and `calc_result` function implementations. These will be marked automatically using predefined test cases with set function inputs and expected return values. It is thus important that you test these functions using the emulator and the STM32CubeIDE first, before submitting the code in the answer boxes. The functions may not make use of global variables or structures or definitions declared outside of the function. You may assume that the stdio library has been included, and that the standard numeric types (i.e. uint8_t, int32_t etc) have been declared. You should not call any HAL_* functions in the two submitted function implementations. (You may however make use of the HAL_* functions outside of the `receive_byte` and `calc_result` functions).

For the complete program, we require that you upload a BIN file of your completed program. Marking will occur automatically based on the BIN file only. The complete project source code does not get submitted, and no marks are based on the project source code.

Die volgende twee vrae vereis dat jy die `receive_byte` en `calc_result` funksie implementerings indien. Hierdie kode inhandigings sal outomaties gemerk word deur gebruik te maak van toetsgevalle met voorafbepaalde intree en verwagte terugkeer resultate. Dit is dus belangrik dat jy die funksies behoorlik toets voor jy indien deur van die emuleerder en STM32CubeIDE gebruik te maak. Die funksie implementerings mag nie gebruik maak van globale veranderlikes, strukture of nommer-tipes wat buite die funksie verklaar moet word nie. Jy mag aanneem dat die stdio biblioteek beskikbaar is, en dat die standaard numeriese tipes (soos uint8_t int32_t ens.) reeds verklaar is. Jy mag nie die HAL_* funksies roep vanuit hierdie twee funksies se implementerings nie. (Jy mag wel die HAL_* funksies gebruik buite die `receive_byte` en `calc_result` funksies).

Vir die volledige program word vereis dat jy die BIN lêer van die voltooide program sal indien. Hierdie vraag sal outomaties gemerk word gebasseer op die BIN lêer alleenlik. Die voltooide projek se volledige bronkode word nie ingedien nie - slegs die bronkode van die 3 funksies in die vorige vrae.

Information

The implementaiton for the `process_input` function is given below. You can use this in your program.

Die implementering van die `process_input` funksie word hieronder gegee. Jy kan dit in jou program gebruik.

```
/**
* @brief Extract the two operands and operator from the input buffer
*
* This function will process the input buffer and extract the two operands (num1 and num2)
* from it as 16-bit signed integers. It will also determine which operation is required
* (+,-,x,/) and return a numeric code based on the operation.
* The operands as well as operation code are returned using pointers, supplied as
* function parameters. The function should return 1 if the input is correctly formatted
* and 0 otherwise.
*
* @param inputbuffer The array containing the input to process. maximum length is 11
* @param ptr_num1 Address where the first extracted number must be stored.
* @param ptr_num2 Address where the second extracted number must be stored.
* @param ptr_opcode Address where the operation must be stored.
*                   1 = addition, 2 = subtraction, 3 = multiplication, 4 = division
*
* @return 1 if the input is correctly formatted, 0 otherwise
*/
```

```
uint8_t process_input(uint8_t* inputbuffer, int16_t* ptr_num1, int16_t* ptr_num2, uint8_t* ptr_opcode)
{
   int16_t num1 = 0;
   int16_t num2 = 0;
   int ix = 0;
   int n1sign = 0;
   if (inputbuffer[0] == '-')
   {
      n1sign = 1;
      ix++;
   }
   while ((inputbuffer[ix] != '+') &&
          (inputbuffer[ix] != '-') &&
          (inputbuffer[ix] != 'x') &&
          (inputbuffer[ix] != '/'))
   {
      if ((inputbuffer[ix] >= '0') && (inputbuffer[ix] <= '9'))
      {
         if (num1 >= 9999)
            return 0;

         num1 *= 10;
         num1 += (inputbuffer[ix] - 48);
      }
      else
         return 0;

      ix++;
   }
   if (n1sign)
      num1 = -num1;

   switch (inputbuffer[ix])
   {
      case '+' : *ptr_opcode = 1; break;
      case '-' : *ptr_opcode = 2; break;
      case 'x' : *ptr_opcode = 3; break;
      case '/' : *ptr_opcode = 4; break;
   }
   ix++;

   while (inputbuffer[ix] != '=')
   {
      if ((inputbuffer[ix] >= '0') && (inputbuffer[ix] <= '9'))
      {
         if (num2 >= 9999)
            return 0;

         num2 *= 10;
         num2 += (inputbuffer[ix] - 48);
      }
      else
         return 0;

      ix++;
    }

    *ptr_num1 = num1;
    *ptr_num2 = num2;

  return 1;
}
```

Question **1**

Incorrect

Marked out of 6.00

Copy your receive_byte function implementation here. Note that the "Check" button for this question will not provide any feedback, and is completely non-functional.

Kopieer jou receive_byte funksie implementering hier. Let wel dat die "Check" knoppie vir hierdie vraag geen terugvoer sal verskaf nie, en vir alle doeleindes nie-funksioneel is.

Answer: (penalty regime: 0 %)

Reset answer

```
uint8_t receive_byte(uint8_t* inputbuffer, uint8_t* ptr_writepos, uint8_t rxbyte)
{
   // pseudocode of what must be implemented:
   // if writepos < 11
   //    . write rxbyte into inputbuffer at index writepos
   //    . increment writepos
   //
   // if rxbyte is a '='
   //    return 1
   // else
   //    return 0
}
```

Question **2**

Incorrect

Marked out of 5.00

Copy your calc_result function implementation here. Note that the "Check" button in this question will not provide any feedback, and is completely non-functional.

Kopieer jou calc_result funksie implementering hier. Let wel dat die "Check" knoppie vir hierdie vraag geen terugvoer sal verskaf nie, en vir alle doeleindes nie-funksioneel is.

Answer: (penalty regime: 0 %)

Reset answer

```
int32_t calc_result(int16_t num1, int16_t num2, uint8_t opcode)
{

}
```

Question **3**

Not answered

Marked out of 10.00

Save and upload your program emulator BIN file for the completed program. Please use your student number for the filename, i.e. 12345678.bin

Stoor die BIN lêer van jou voltooide program vanaf die emuleerder en laai dit hier op. Gebruik asb. jou studentenommer vir die lêernaam, bv. 12345678.bin

Question **4**

Complete

Marked out of 8.00

Choose True or False for the following questions:	Kies Waar of Vals vir die volgende vrae:
The ARM Cortex M4 that was used in this module is a microcontroller.	Die ARM Cortex M4 wat in hierdie module gebruik was is 'n mikrobeheerder.

False/Vals — False

The push-buttons and LEDs are part of the microcontroller.	Die knoppies en LEDs is deel van die mikrobeheerder.

False/Vals — False

The microcontroller usually includes memory on the same chip as the CPU.	Die mikrobeheerder sluit gewoonlik geheue in op dieselfde skyfie ("chip") as die CPU.

True/Waar — true

The STM32F411VET implements the Von Neuman Machine architecture.	Die STM32F411VET implementeer die Von Neuman Masjien argitektuur.

False/Vals — False

If you write a program only using assembly language and save these files as .s files, then these files can be copied directly to the Flash memory of the microcontroller in order to run the code.	As jy 'n program skryf wat slegs saamsteltaal gebruik en hierdie as .s-lêers stoor, kan hierdie lêers direk na die Flash-geheue van die mikrobeheerder gekopieër word om die kode uit te voer.

False/Vals — False

A PCB can still fully function without a silkscreen or soldermask.	'n PCB kan steeds volledig funksioneer sonder 'n syskerm of soldeermasker.

False/Vals — Yes

The DMA controller can execute code.	Die DMA beheerder kan kode uitvoer.

False/Vals — False

The MCU can execute code that is copied into SRAM.	Die mikrobeheerder kan kode uitvoer wat na SRAM gekopieer is.

True/Waar — True

Question **5**

Complete

Marked out of 3.00

The development board that was discussed in this module (STM32F411 Discovery Kit) only has a 8 MHz oscillator on the development board, but on the specifications of the board the packaging says it runs at 100 MHz. Is this a mistake? If it is a mistake, explain how they could have accidentally made this mistake. If it is not a mistake, explain how it can be true.	Die ontwikkelingsbord wat in hierdie module bespreek is (STM32F411 Discovery Kit) het slegs 'n 8 MHz-ossillator op die ontwikkelingsbord, maar volgens die spesifikasies van die bord sê die verpakking dat dit op 100 MHz werk. Is dit 'n fout? As dit 'n fout is, verduidelik hoe hulle per ongeluk hierdie fout kon maak. As dit nie 'n fout is nie, verduidelik hoekom dit waar kan wees.

Its not mistake.
The frequency cannot be changed, it is fixed at 8MHz if is satistfies that we have SB14,SB16 OPEN, R25 removed and R66 soldered

Not a mistake. We can use a phase locked loop to multiply base clock frequency to desired output frequency. (L12)&pg90

Question **6**

Complete

Marked out of 4.00

SRAM is much faster than Flash memory. Assuming that cost was not a factor, why do we have both on the STM32F411 microcontroller and what is mainly stored in each?	SRAM is baie vinniger as Flash-geheue. Gestel dat die koste nie 'n faktor was nie, waarom het ons albei op die STM32F411-mikrobeheerder en wat word hoofsaaklik in elk gestoor?

Because memory types cannot be accessed directly by CPU

SRAM is static since the store bits that do not need to be transferred. It is used for cache memory. It stores volatile memory

Flash memory contains NAND and NOR. It stores non-volatile memory and vector table

SRAM uses more parts per memory cell (therefore smaller) and because of this it is only used to store frequently used data

Question **7**

Complete

Marked out of 4.00

You have a four input buttons that are connected to four GPIO pins. These pins are set up to trigger external interrupts on the rising edge using EXTI0 to EXTI3 with the priorities set as shown in the table below. Each ISR, that handles the interrupts, take 5 seconds to complete its function before control returns to the main program. (Note that wrong answers are negatively marked, **so don't guess**)	Jy het vier intree-knoppies wat aan vier GPIO-penne gekoppel is. Hierdie penne is opgestel om eksterne onderbrekings op die stygende flank te aktiveer met behulp van EXTI0 tot EXTI3 met die prioriteite opgestel soos in die tabel hieronder getoon. Elke ISR, wat die onderbrekings hanteer, neem vyf sekondes om die funksie te voltooi voordat die beheer na die hoofprogram terugkeer. (Let daarop dat verkeerde antwoorde negatief gemerk word, **dus moenie raai nie**)

Button Number	Interrupt	Interrupt priority (NVIC_IPR)
1	EXTI0	6
2	EXTI1	5
3	EXTI2	5
4	EXTI3	4

If the ISR for Button 1 is busy executing, which buttons will be able interrupt the current ISR (you can select multiple options).	As die ISR vir "Button 1" besig is om uit te voer, watter knoppies kan die huidige ISR onderbreek (jy kan meer as een opsie kies).

☐ Button 1 ☑ Button 2 ☑ Button 3 ☑ Buttons 4 ☐ None of the buttons Correct

If the ISR for Button 2 is busy executing, which buttons will be able interrupt the current ISR (you can select multiple options).	As die ISR vir "Button 2" besig is om uit te voer, watter knoppies kan die huidige ISR onderbreek (jy kan meer as een opsie kies).

☐ Button 1 ☐ Button 2 ☐ Button 3 ☑ Buttons 4 ☐ None of the buttons Correct

If the ISR for Button 4 is busy executing, which buttons will be able interrupt the current ISR (you can select multiple options).	As die ISR vir "Button 4" besig is om uit te voer, watter knoppies kan die huidige ISR onderbreek (jy kan meer as een opsie kies).

☐ Button 1 ☐ Button 2 ☐ Button 3 ☐ Buttons 4 ☑ None of the buttons Correct

At which memory address does the ISR for Button 2 start.	By watter geheue adres begin die ISR vir "Button 2".

○ 0x40 ○ 0x58 ○ 0x5C ◉ 0x60 ○ 0x1001 ○ Not enough information Correct

Question **8**

Complete

Marked out of 3.00

A first year student finds the code for the ISRs in the previous question and tries to improve it, but in the process accidently causes the ISR for Button 1 to go into an infinite loop during its execution. Luckily the STM32F411 microcontroller includes some functionality that can recover from unexpected infinite loops. What is this functionality and briefly explain how it works for 2 marks?	'n Eerstejaarstudent vind die kode vir die ISR's in die vorige vraag en probeer dit verbeter, maar veroorsaak dat die ISR vir Button 1 per ongeluk in 'n oneindige lus ingaan tydens die uitvoering daarvan. Gelukkig bevat die STM32F411 mikrobeheerder sekere funksionaliteit wat kan herstel van onverwagte oneindige lusse. Wat is hierdie funksionaliteit en verduidelik kortliks hoe dit werk vir 2 punte?

NVIC

WATCHDOG TIMER - To prevent the timer from reaching zero, write to a register to kick the watchdog - causes counter register to start again with reload value.

Question **9**

Complete

Marked out of 8.00

Lighthouses in fairly close proximity to each other have different flash patterns to allow the mariner to identify their location. You need to build a new electronic lighthouse that would have a 0.5 second flash of light every 4 seconds (therefore **off for 3.5 seconds and on for 0.5 seconds**). You decide to use a timer of your STM32F411 microcontroller to achieve this. Generate a PWM signal using TIM2 in PWM Mode 1 in center-aligned mode. You must therefore calculate values for the clock frequency, TIM2_PSC, TIM2_ARR and TIM2_CRR that will implement this flash pattern. In each of the following four cases you are given three of the four values and you must calculate the missing value.	Vuurtorings in 'n wat naby aan mekaar is, het verskillende flitspatrone om die seevaarder in staat te stel om hul ligging te identifiseer. Jy moet 'n nuwe elektroniese vuurtoring bou wat elke 4 sekondes 'n flits van 0.5 sekonde sal hê (dus vir **3.5 sekondes af en dan vir 0.5 sekondes aan**). Jy besluit om 'n timer van jou STM32F411 mikrobeheerder te gebruik om dit te bereik. Genereer 'n PWM-sein met behulp van TIM2 in PWM-modus 1 in die "center-aligned mode". Jy moet dus waardes bereken vir die klokfrekwensie, TIM2_PSC, TIM2_ARR en TIM2_CRR wat hierdie vereiste flitspatrool sal implementeer. In elk van die volgende vier gevalle word drie van die vier waardes gegee en jy moet die ontbrekende waarde bereken.

TIM2_PSC = 999
TIM2_ARR = 32000
TIM2_CRR = 4000

TIM2 clock frequency (MHz) = 0.016 16 Mhz

TIM2 clock frequency (MHz) = 90 MHz
TIM2_ARR = 36000
TIM2_CRR = 4500

T = (PSC)(ARR)/Fclk

TIM2_PSC = 2498.9 5000

IF CENTRE ALLIGNED TIMES BY 2

TIM2 clock frequency (MHz) = 55 MHz
TIM2_PSC = 4999
TIM2_CRR = 2750

TIM2_ARR = 10999 22000

TIM2 clock frequency (MHz) = 68 MHz
TIM2_PSC = 4 999
TIM2_ARR= 27200

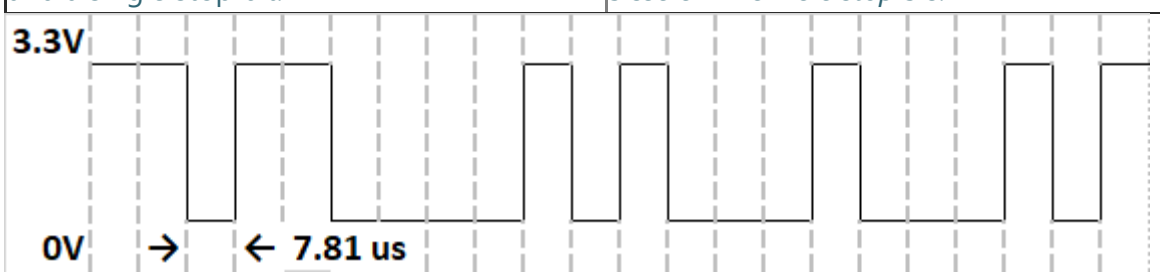
TIM2_CRR =

Question **10**

Complete

Marked out of 6.00

A UART signal is plotted below. The UART is configured with no parity bits, 8 data bits and a single stop bit.	*n UART sein word hieronder vertoon. The UART is opgestel vir geen pariteit, 8 data bisse en 'n enkele stop bis.*



Determine the closest standard baudrate of the channel	*Bepaal die naaste standaard baud tempo van die kanaal*

4800 baud

What are the two ASCII characters that are being transmitted?	*Wat is die twee ASCII karakters wat oorgedra word?*

1st character / *1ste karakter* : A C

2nd character / *2de karakter* : Z D

Question **11**

Complete

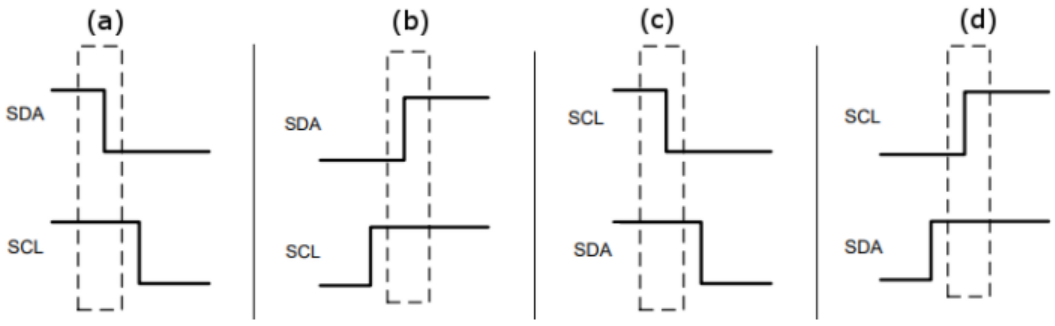
Marked out of 3.00

What is the normal state of the I2C SCL and SDA lines when no communication is taking place?	Wat is die normale toestand van die I2C SCL en SDA lyne as daar geen kommunikasie plaasvind nie?

SCL is high while SDA is held low / SCL is hoog terwyl SDA laag gehou word

False, both high

Identify the I2C start and stop conditions from the following image.	Identifiseer die I2C "start" en "stop" kondisies in die volgende diagram.



Start: (a) [1]

true

Stop: (b) [1]

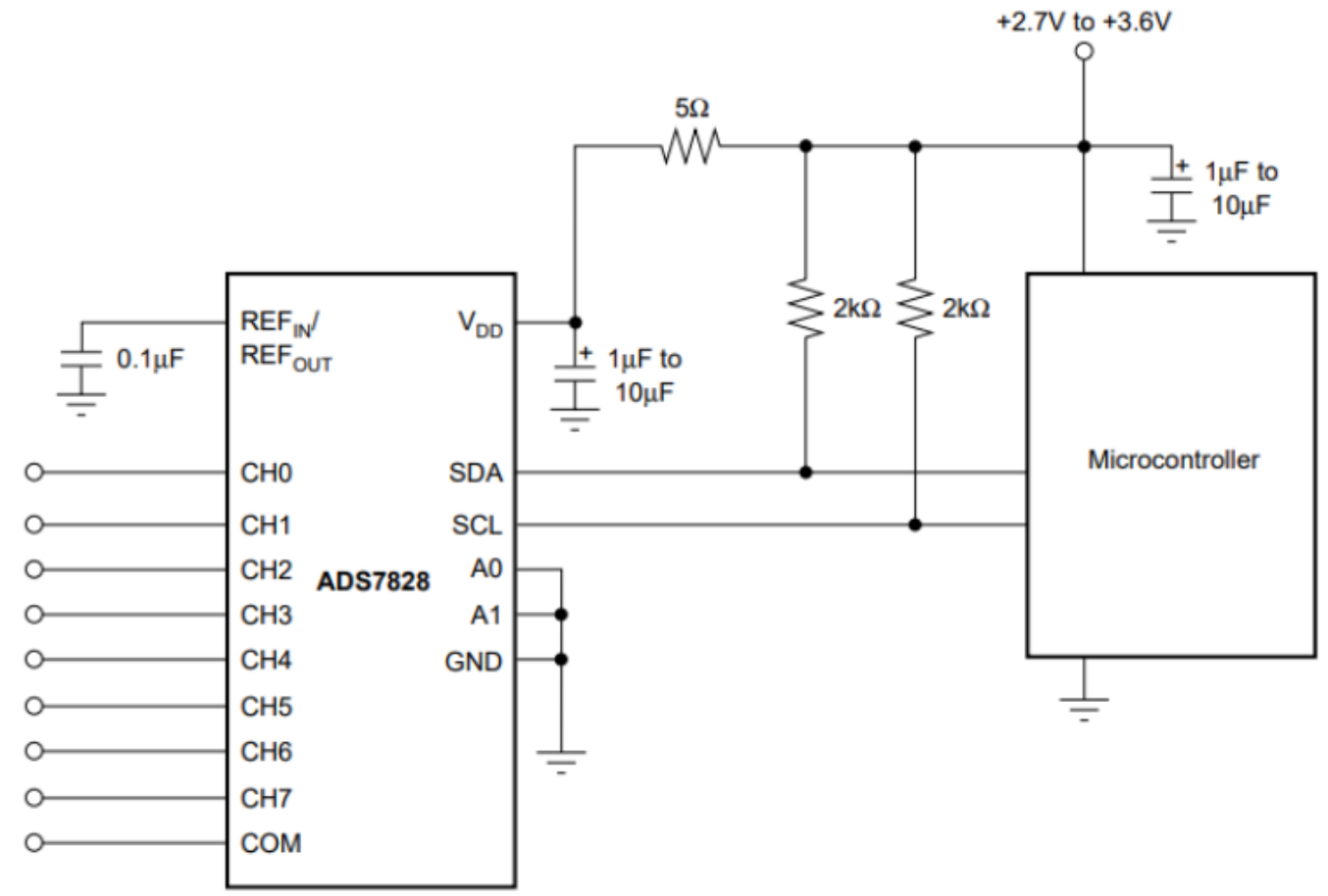
true

Question **12**

Complete

Marked out of 6.00

A project you are working on has a STM32 microcontroller connected to an external ADC, the ADS7828. The latter device uses I2C communication to the microcontroller. The connection diagram is as below.	'n Projek waaraan jy werk maak gebruik van 'n STM32 verwerker wat verbind is aan 'n eksterne ADC, 'n ADS7828 toestel. Die kommunikasie tussen die mikroverwerker en die ADC maak gebruik van I2C soos in die diagram hieronder.



The relevant portion of the datasheet regarding the I2C slave address is repeated below	Die relevante deel van die datavel met betrekking tot die I2C adres word hieronder herhaal

ADDRESS BYTE

MSB	6	5	4	3	2	1	LSB
1	0	0	1	0	A1	A0	R/$\overline{W}$

The address byte is the first byte received following the START condition from the master device. The first five bits (MSBs) of the slave address are factory pre-set to 10010. The next two bits of the address byte are the device select bits, A1 and A0. Input pins (A1-A0) on the ADS7828 determine these two bits of the device address

What is the 7-bit slave address for the device in the given circuit diagram? (answer in hexadecimal)	Wat is die 7-bis slaaf adres van die ADS7828 gegee die boonste stroombaandiagram? (Antwoord in heksadesimaal)

0x 48 Correct

What is the maximum number of ADS7828 devices that can be connected to the same I2C bus without causing an address clash?	Wat is die maksimum aantal ADS7828 toestelle wat aan dieselfde I2C bus verbind kan word sonder om adres botsings te veroorsaak?

4

Complete the following function arguments for the given address configuration	Voltooi die volgende funksie argumente vir die gegewe adres konfigurasie

HAL_I2C_Master_Transmit(&hi2c1, 0x 48 , data, 1, 10);

HAL_I2C_Master_Receive(&hi2c1, 0x 48 , data, 6, 10);

For the HAL_I2C function calls above, what does the 4th function argument represent?	Vir die HAL_I2C funksie hierbo, Wat verteenwoordig die 4de funksie argument?

The amount of bytes to transmit (or receive)/Die aantal grepe om te stuur (of te ontvang) true

What must the data type of the 'data' argument be?	Wat moet die datatipe van die 'data' argument wees?

uint8_t *

false - not a pointer so -> uint_8

Question **13**

Complete

Marked out of 6.00

An analogue-to-digital converter (ADC) on a microprocessor is used to sample the voltage on one of the input pins. The ADC has 12-bit resolution and a reference voltage (VREF) of 3.3V. If the voltage on the ADC input pin is 1V, what is the expected sampled A/D value?	'n Analoog-na-digitaal omsetter (ADC) van 'n mikroverwerker word gebruik om die spanning op een van die intree penne te meet. Die omsetter het 12-bis resolusie en 'n verwysings spanning (VREF) van 3.3V. As die spanning op die ADC intree 1V is, wat is die verwagte gemonsterde A/D waarde?

1240

TRUE

What is the smallest non-zero voltage that can be measured by the ADC (in µV units, rounded to the nearest integer)?	Wat is die kleinste nie-nul spanning wat deur die analoog na digitaal omsetter gemeet kan word (in µV eenhede, afgerond na die naaste heelgetal)?

806 µV

Correct

If the ADC is driven by a 10MHz clock signal, and it takes 11.5 clock cycles to perform one conversion, what is the maximum conversion rate (in samples per second, rounded to the nearest integer)?	Indien die analoog-na-digitaal omsetter deur 'n 10MHz kloksein gedryf word en dit neem 11.5 kloksiklusse om 'n enkele omskakeling te doen, wat is die maksimum monsterfrekwensie (in monsters per sekonde, afgerond na die naaste heelgetal)?

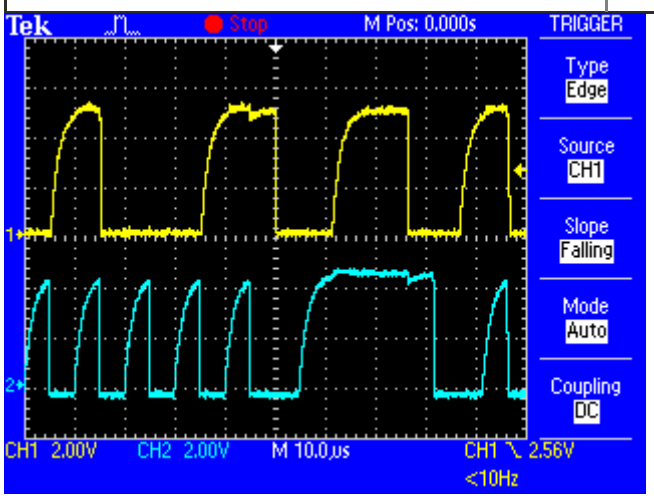
869565 sps

Question **14**

Complete

Marked out of 4.00

After measuring the I2C clock and data signals for a device you are working on using an oscilloscope, you obtain the following result:	Jy kry die volgende resultaat as jy die I2C klok en data seine van 'n toestel waaraan jy werk meet op 'n ossiloskoop:



What can be done to improve the signal rise time?	Wat kan gedoen word om te verseker dat die seine vinniger sal styg?

Increase the pull-up resistance / Verhoog die optrekweerstand

decrease

Which of the following situations would likely require impedance matching?	Watter van die volgende situasies sal waarskynlik impedansie gelykstelling benodig?
10Gbit/s ethernet over 100m length	10Gbit/s "ethernet" (rekenaar netwerk) oor 'n 100m afstand

Yes / Ja TRUE

100kHz I2C bus between MCU and sensor on the same PCB	100kHz I2C bus tussen mikrobeheerder en sensor op dieselfde gedrukte stroombaan bord

No / Nee TRUE

A microcontroller that switches a heater on using a GPIO output pin in a 10s control loop. The power switch and MCU are on the same PCB	'n Mikrobeheerder wat 'n verhitter aanskakel deur 'n GPIO pen in 'n 10s beheerlus. Die kragskakelaar en mikrobeheerder is op dieselfde gedrukte stroombaan bord

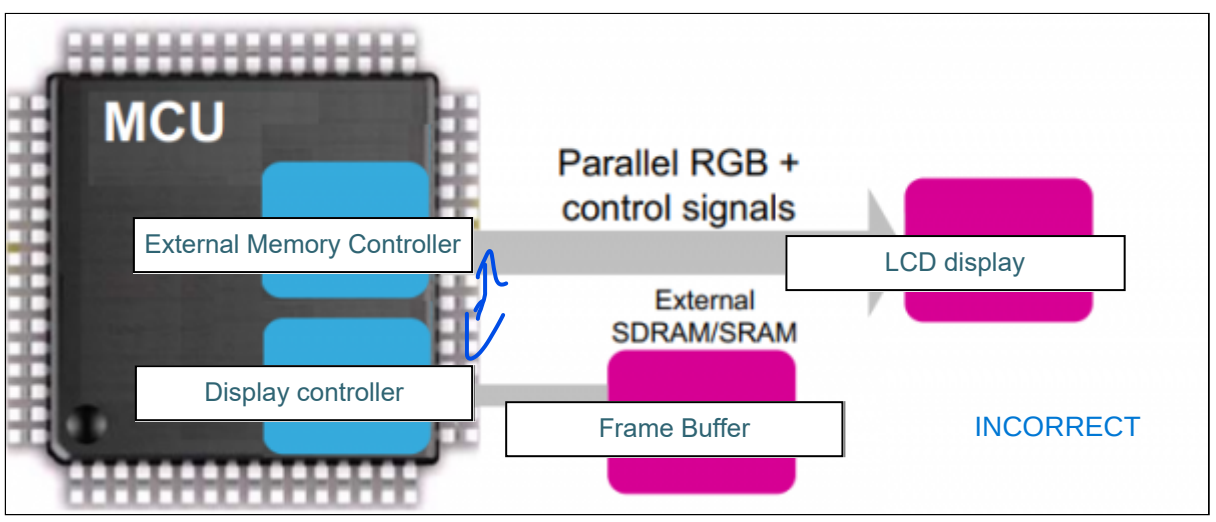
No / Nee ?

Question **15**

Complete

Marked out of 4.00

Complete the block diagram of a MCU that interfaces with a display panel, and makes use of external SRAM for the frame buffer, by dragging the labels to their correct locations.	Voltooi die blok diagram van 'n mikrobeheerder wat met 'n vertoonpaneel verbind is, en gebruik maak van eksterne SRAM vir die vertoon geheue, deur die kentekens te skuif na die regte plekke op die diaigram.



Question **16**

Complete

Marked out of 10.00

An IoT number plate recognition system makes use of a camera and image processing to detect vehicle number plates. The extracted registration numbers are sent to a server along with a timestamp. The system has the following characteristics:

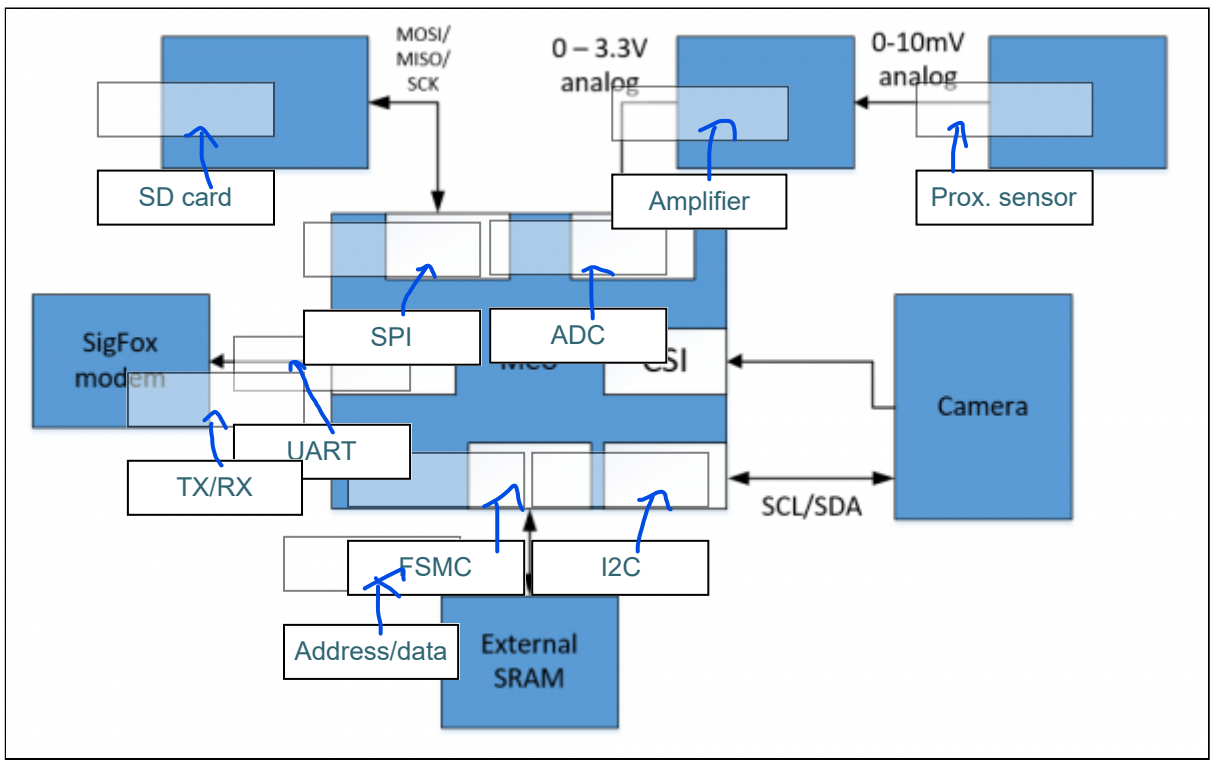
- The system makes use of a microcontroller to perform the image processing and server communications.
- The microcontroller interfaces to the camera using a Camera Serial Interface (CSI)
- The camera also has an I2C interface which is used for control signals, and to adjust gain and exposure
- The system makes use of a proximity sensor sensor to detect passing vehicles. The camera is triggered to capture a photo only if a passing vehicle is sensed
- The proximity sensor has an analogue output in the range 0 to 10mV, corresponding to a vehicle distance of 10cm to 10m. The microcontroller should take a photo when the distance is 3m
- The microcontroller makes use of external SRAM because the internal SRAM is not enough to store a full camera frame.
- The microcontroller is an STM32 range microcontroller and contains a Flexible Static Memory Controller (FSMC)
- The device also makes use of an SD card to store photos and logged number plates.
- Communication to the server makes use of the SigFox network. The modem has a UART interface.

Complete the block diagram of the system below, by dragging the labels of blocks, interfaces and signals into the correct location

'n Internet-van-dinge nommerplaat herkenning stelsel maak gebruik van 'n kamera en beeldverweking om voertuie se nommerplate te herken en die registrasie nommer te onttrek. Die registrasienommer word dan na 'n bediener gestuur saam met die huidige tyd. Die stelsel het die volgende karakteristieke:

- Die stelsel maak gebruik van 'n mikrobeheerder om die beeldverwerking en kommunikasie na die bediener te behartig.
- Die mikrobeheerder verbind aan die kamera deur 'n Camera Serial Interface (CSI) koppelvlak.
- Die kamera het ook 'n I2C koppelvlak vir beheer bevele en om die kamera se integrasietyd en aanwins te stel.
- Die stelsel maak gebruik van 'n "nabyheidsensor" (proximity sensor) om die afstand na aankomende voertuie te meet.
- Die nabyheidsensor het 'n analoog spanningsuittree van 0 tot 10mV, wat ooreenstem met 'n voertiugafstand van 10cm na 10m. Die mikrobeheerder moet die kamera beveel om 'n foto te neem as die nabyheidsensor 'n afstand van 3m registreer.
- Die mikrobeheerder maak gebruik van eksterne SRAM omdat die interne geheue nie voldoende is om 'n volledige kamera beeld te stoor nie.
- Die mikrobeheerder is een van die STM32 reeks mikrobeheerders en bevat 'n "Flexible Static Memory Controller" (FSMC).
- Die toestel maak ook gebruik van 'n SD kaart om fotos en registrasie nommers te stoor.
- Kommunikasie na die bediener maak gebruik van die SigFox netwerk. Die modem het 'n UART koppelvlak.

Voltooi die blokdiagram van die stelsel hieronder deur die kentekens van blok name, koppelvlakke en seine te skuif na die korrekte plekke.



◄ libuart example static library

Jump to...

Appendix A - Digital System Implementation ►